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Signal Processing Methods for Artifact Removal in TMS-EEG Data

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List of acronyms

BSS Blind Source Separation

DWT Discrete Wavelet Transform

EEG Electroencephalography

EMD Empirical Mode Decomposition

EOG Electrooculography

ICA Independent Component Analysis

IMFs Intrinsic Mode Functions

MAA Mean Absolute Amplitude

OAs Ocular Artifacts

PCA Principal Component Analysis

PSD Power Spectral Density

RMSE Root Mean Square Error

SIR Source-Informed Reconstruction

SOUND Source-Utilized Noise-Discarding algorithm

SSP Signal Space Projection

TEPs TMS-Evoked Potentials

TMS Transcranial Magnetic Stimulation

WD Wavelet Decomposition

1
Introduction

The Electroencephalography (EEG) is a fundamental tool for studying brain activity. Its combination with Transcranial Magnetic Stimulation (TMS) has opened new research possibilities. However, EEG data is subject to the presence of artifacts (i.e. noise), and the integration with TMS brings even more noise to deal with. We define as an artifact any non-central nervous system signal recorded by the EEG, introduced by the subject himself or by an external source [7]. Artifacts can compromise the analysis and interpretation of TMS-evoked potentials (TEPs). Due to its impulsive and high-intensity nature, TMS can introduce various artifacts (muscular, ocular, electromagnetic, sensorial) that can distort the signal of interest. This work is focused on the comparison of different methods for artifact removal in TMS-EEG data. In particular the implementation and analysis of two consolidated pipelines: the Two-Step Fast Independent Component Analysis [24] and the Source-Utilized Noise-Discarding algorithm incorporating Signal Space Projection [17, 18]), and the examination of alternative methods for ocular artifact removal, like Wavelet Decomposition (WD) and Empirical Mode Decomposition (EMD). The study was performed using a dataset provided by Romero Lauro et al. [13]. In summary, the target of this work is to explore the most common strategies and investigate alternative possibilities in the TMS-EEG preprocessing domain.

This thesis is part of a broader interdisciplinary effort carried out in collaboration with the Department of Psychology, specifically with the research group led by Prof. Romero Lauro. In parallel, a related project was conducted by Diego Corna, a Master's student in Artificial Intelligence, who explored the application of Deep Learning techniques to TMS-EEG data.

2
Background

2.1 Brain overview

The **brain** is the organ that controls the nervous system. It is made of **neurons** (which are composed of a cell body, the dendrites, and the axon) and by **support cells** (glia). Neurons are organized within the brain to form white matter and gray matter. The former consists of axons and support cells, while the latter is composed of neuronal cell bodies. The **cerebral cortex** is the outer part of the brain, it is around 3 mm thick and it is the area of the brain mostly affected by TMS. It is organized into two hemispheres, left and right, and consists of two folded sheets of gray matter.
The cortex is divided into 4 regions called lobes (Figure 2.1):

- **Frontal lobe**: executive functions and motor control;
- **Parietal lobe**: sensation and spatial reasoning;
- **Temporal lobe**: hearing and memory;
- **Occipital lobe**: brain visual processor.

The brain's electrical activity generates rhythmic patterns known as brainwaves, which can be measured using electroencephalography (EEG). These oscillations reflect the synchronized activity of large populations of neurons and are associated with different states of consciousness and cognitive processes. Brainwaves are characterized by their frequency,

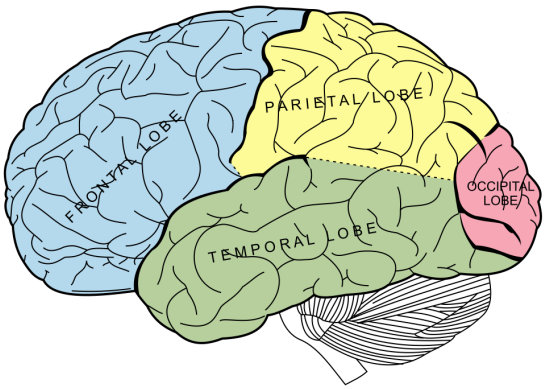


Figure 2.1 The cerebral cortex divided in its four lobes. *Source: Gray's Anatomy via Wikimedia Commons.*

measured in Hertz (Hz), and are classified into several distinct bands (Table 2.1). Most of the healthy brain activity is below 50 Hz.

Brainwave Type	**Frequency Range**	**Associated States and Functions**
Delta (δ)	0.5-4 Hz	Deep sleep, newborns
Theta (θ)	4-8 Hz	Emotional stress, drowsiness
Alpha (α)	8-12 Hz	Relaxed wakefulness state
Mu (μ)	8-12 Hz	Motor cortex functionality
Beta (β)	12-30 Hz	Alert state, active thinking/attention, anxiety
Gamma (γ)	$\geq$ 30 Hz	Intense brain activity

Table 2.1 Brain waves frequencies classified into bands.

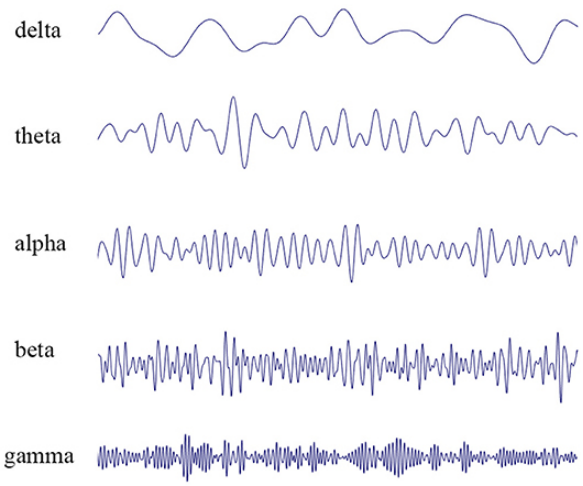


Figure 2.2 Brain signal decomposed in delta, theta, alpha, beta and gamma bands [27].

2.2 TMS-EEG

Combining Transcranial Magnetic Stimulation (TMS) with Electroencephalography (EEG) offers a powerful approach to study the brain in ways that neither technique can achieve alone.

The **Transcranial Magnetic Stimulation (TMS)** is a non-invasive method used to stimulate the brain. It works by sending a brief, strong electric current through a coil placed near the scalp. This creates a magnetic field that passes through the skull and induces an electric field in the brain, which can activate neurons in the targeted area. TMS is most effective at stimulating the outer layers of the brain (Section 2.1), but depending on how strong the stimulation is and how it is directed, TMS can cause neurons to fire and send signals to other parts of the brain. This allows researchers to study how different areas of the brain are connected and interact with each other. The way neurons respond depends on their shape, position, and other properties [8, 10].

The **Electroencephalography (EEG)** is the most commonly used brain imaging tool to measure the effects of TMS stimulation. It is a non-invasive method for recording brain activity, offering excellent temporal resolution on the scale of milliseconds. EEG measures voltage differences between pairs of electrodes placed on the scalp, capturing the brain's electrical activity in real time. The signals recorded by EEG mainly reflect postsynaptic potentials, the electrical changes that occur when neurons receive input from other neurons, rather than action potentials, which are the brief signals used by neurons to send messages. Postsynaptic potentials are longer in duration and more likely to occur in synchrony across many neurons, which makes them more detectable at the scalp level. When thousands of neurons with similar orientation in the cortex become active at the same time, their electrical signals can add up and be measured as EEG signals. Because of this, EEG is especially sensitive to large-scale, synchronized brain activity and is a valuable tool for studying brain function and dynamics over time [3, 10].

This combination is particularly advantageous considering that it can reveal how excitable or connected different brain regions are. By analyzing EEG signals that occur after a TMS pulse, known as TMS-evoked potentials (TEPs) or brain oscillations, researchers can gain valuable information about the function of both stimulated and connected regions [10]. Figure 2.3 shows the seqeuence of events of TMS-EEG data acquisition.

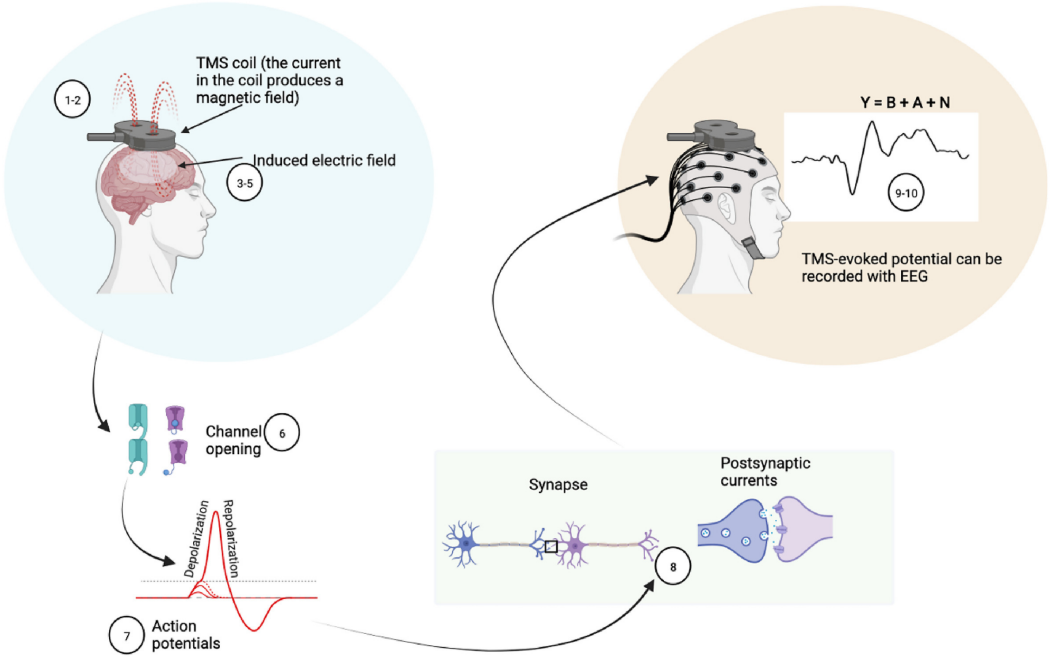


Figure 2.3 Sequence of events triggered by a TMS pulse: (1–2) A strong current in the TMS coil generates a brief, high-intensity magnetic field; (3) The changing magnetic field induces an electric field in the brain; (4–5) This causes current flow in the tissue and local membrane depolarization; (6–7) Voltage-gated ion channels open, leading to the generation of action potentials in axons; (8–9) Neurotransmitter release in the synapse generates postsynaptic potentials, which may trigger further action potentials across the network; (10) These network responses produce electric potentials that reach the scalp and are recorded by EEG. The EEG signal reflects a combination of brain activity (B), artifacts (A), and noise (N), modeled as: $Y = B + A + N$. Figure created with BioRender.com. Source: Hernandez-Pavon et al. [10]

2.3 TMS-EEG Artifacts

An artifact is an unwanted signal or noise that contaminates the true brain electrical activity. TMS-EEG artifacts can be categorized into two main categories, based on their origin, physiological and non-physiological artifacts. The former are caused by the body's activity, while the latter by mechanical or electric events. The TMS pulse can induce various artifacts that can be strictly temporally related to the TMS pulse. They are referred to as time-locked.

2.3.1 Non-physiological artifacts

Pulse artifact or electromagnetic artifact

This is the largest artifact generated by the TMS pulse. It is of electromagnetic nature and is produced by the electromotive force induced in the loops formed by EEG electrode leads. It can be up to several volts, causing two main problems: (a) the masking of the brain signals, (b) the saturation of the EEG amplifier [11, 28].

Even though the brain signal during the pulse cannot be recovered, these problems can be mitigated. For example, this can be achieved by using an EEG amplifier designed to handle pulse artifacts, or by minimizing its duration with an adequate sampling frequency and a sufficiently high cutoff frequency for the anti-aliasing low-pass filter [10].

TMS recharge artifacts

This artifact is produced when the capacitors, which store the electric charge required for TMS, are recharged. The recharge artifact can look like a spike, an abrupt signal jump, an exponential decay, or a waning high-frequency discharge [11, 28].

The solution is to manually adjust the capacitor recharge timing in order to delay the artifact outside of the time window of interest [10].

Decay artifact

The electrode-skin interface can be polarized by electrical currents between the electrolyte gel and the recording electrode. This typically results in an exponentially decaying charge, where the decaying current is proportional to the remaining polarization voltage [11, 28].

The solution to minimize the decay artifact is to reduce the impedance of the skin-electrode interface and use non-polarizable electrodes and electrolyte materials [10].

Electrode motion artifact

This artifact is generated by the movement of the electrode against the electrolyte and is often caused by: (a) vibration of the TMS coil [15], (b) muscle twitch or head movements induced by the TMS pulse, (c) the coil or the operator touching the electrode or (d) movement-related skin stretching. The artifacts generated by the pulse can be seen as they usually occur within the first $\sim$ 10 ms after the pulse [28].

There are several practical solutions to this problem: (a) adding a thin layer of foam between the coil and the electrodes, (b) wrapping the EEG cap with a layer of cellophane and/or an elastic net bandage and (c) using 3D printable spacers to separate the coil from the electrodes [10, 15].

2.3.2 Physiological artifacts

Ocular artifacts

This artifact is a consequence of the electrical field generated by the eyes; in fact, the eye acts as a strong dipole, with the front of the eye positively charged and the back of the eye negatively charged. Thus, any eye movement can slightly modulate the dipole, causing artifacts in the EEG. In particular, TMS can cause a startle reflex due to the coil click [15, 11].

The solution is to train the subject to reduce blink artifacts. In contrast, when eye movements are spontaneous, it is easier to remove the artifact using offline signal processing techniques [10].

Cranial muscle artifacts

These artifacts can be induced by the TMS pulse when muscles innervating the head/face are stimulated and can severely contaminate the EEG. The most likely muscles to be activated are neck, facial, frontal, temporal, and masseter muscles, depending on the placement of the TMS coil. It should also be noted that muscle contractions can cause electrode movement, which can cause the electrode motion artifact [15, 11, 28].

The solution is to move the location or change the orientation of the TMS coil. If this is not possible, reducing the TMS intensity or using a smaller, more focused coil is suggested. However, if muscle artifacts cannot be avoided during TMS-EEG data acquisition, certain artifact removal methods can be used offline to remove or suppress the muscle artifacts [10].

Auditory artifacts

These artifacts are caused by the magnetic field generated by the TMS coil producing strong forces on the currents in the coil windings, resulting in a loud clicking sound. This sound has been shown to produce an auditory evoked potential [15, 11, 28].

The solution is to combine noise masking with hearing protection and to place a piece of foam between the coil and the scalp to reduce bone conduction to the inner ear [10].

Somatosensory responses

This artifact can be caused by the TMS having that the somatosensory system may be stimulated by coil vibration, activation of local peripheral sensory axons, activation of the cranial sensory nerve, and cranial muscle twitch [15, 11, 28].

There is no agreed solution to this unwanted signal, as the best strategy depends on the aim of the study [10].

2.3.3 Other artifacts

Artifacts unrelated to TMS

In addition to TMS-induced artifacts, there are other events that can interfere with EEG recordings. These include electrical interference, which can be caused by electronic devices or a poorly grounded laboratory. Physiological artifacts such as electrocardiographic and respiratory signals, tonic and phasic muscle activity, spontaneous movements (including blinking and swallowing), and sweating can also contaminate EEG recordings.

The solution to these artifacts can be found by using electrically shielded rooms and offline filtering, the latter only after the TMS artifacts have been removed [10].

Filtering artifacts

Typically, the filters applied to the EEG can cause ringing, which can be unintentionally interpreted as brain-generated oscillations.

The solution is to avoid unnecessary low-pass filtering, and to increase the cut-off of the anti-aliasing low-pass filter by using appropriate sampling frequencies. Note that any downsampling requires anti-aliasing low-pass filtering. Therefore, downsampling should be avoided before the abrupt high amplitude artifacts are removed from the data [10].

3

State of the art

3.1 Common preprocessing steps

This section explores the most common preprocessing strategies applied to TMS-EEG data to deal with noise and artifacts.

Epoching data around the TMS pulse

This process consists of extracting specific time windows (called "epochs") from a continuous EEG signal, in this case time-locked to the TMS pulse.

For time domain analysis, we need sufficiently long signal segments before the pulse to allow baseline correction and after the pulse to capture the TEPs, which can last up to 400-500 ms after the TMS pulse.

For frequency domain analysis, we need sufficiently long segments to allow sliding-window time-frequency decomposition. The exact length of the segments depends on the frequency of interest and the analysis parameters. [10]

Removing bad channels/trials

This strategy consists of removing the entire channel in the event of interruption during recording, poor electrode contact, ongoing muscle activity, or contact with the TMS coil.

The selection of the channel to be removed can be done manually by the operator or automatically using a statistical approach to identify artifactual data for removal.

Removing noisy data is a highly effective strategy, but care should be taken not to lose too much data that could have been preserved using other artifact removal approaches. [10]

Removing and interpolating the TMS pulse artifact

The most common approach to deal with the TMS pulse artifact is to remove and replace the affected data. The time window to be interpolated depends on the ECG system and configuration. To avoid additional artifacts, it is crucial to replace the removed data with cubic interpolation rather than linear interpolation [10, 24, 19].

Re-referencing the data

The re-referencing of the data is a crucial step to allow a topographical interpretation of the signal. The process consists of re-referencing the EEG data, usually recorded against a single reference electrode, to the common average of all cephalic or non-cephalic electrodes. The use of a common average reference is also useful for comparing data between laboratories, since the positioning of the physical reference may vary [10].

Baseline correction

Baseline correction consists of subtracting a given value from all data points to center the voltage of each electrode around a common reference value. The value can be calculated by taking the average voltage value of the baseline period before the pulse. Baseline correction is an effective strategy for removing the offset caused by factors such as skin hydration and static charges in the electrodes. This step should only be performed after large amplitude artifacts have been removed as they may affect the mean value [10].

Dealing with large amplitude artifacts

High-amplitude artifacts can have a negative impact on temporal filters by producing ringing artifacts, and on blind source separation approaches such as independent component analysis (ICA) by biasing the spatial weights of neural components. Several approaches have been developed to suppress electrode polarization, decay, and TMS-evoked muscle artifacts and to recover the underlying neural signals [9].

One approach is to fit a model representing the artifact to the signal and then subtract the best fit of the model from the data.

Another approach is to use blind source separation algorithms such as ICA or PCA to separate the EEG signal into different components based on temporal or statistical relation-

ships within the data. The problem with ICA and PCA is that we can't really measure the effectiveness of these approaches [10].

Dealing with peripheral evoked potentials in the TMS-EEG

PEPs include auditory and somatosensory evoked responses. Dealing with these EPs can be challenging because they may overlap with TEPs.

There are several approaches that can be followed, including subtracting or regressing a sensory control condition from the TEPs [6], removing components representing PEPs using ICA (at least for the auditory component, [25, 26]), and using a variant of SSP-SIR with a sensory control condition [1, 10].

Temporal filtering

Temporal filters, such as low-pass, band-pass, and band-stop filters, should only be used after TMS pulse, decay, and muscle artifacts have been removed [24, 4, 9]. In general, the use of standard temporal filters to remove TMS-induced artifacts is not recommended. [10].

Interpolate removed channels

When bad channels are removed, a common practice is to interpolate them. Some approaches consist of using spline interpolation of surrounding channels or related methods.

Should be noted that removing or replacing bad channels reduces the dimensionality of the data, which could impact further analysis like ICA and source analysis. [10].

Averaging across trials

After TMS stimulation, the EEG data from each trial is time-locked to the moment of stimulation. To create the final averaged waveform, the voltage values at each time point are summed across all trials and then divided by the total number of trials. This averaging process reduces random noise while preserving the consistent neural responses evoked by TMS [10].

Downsampling the data

TMS-EEG data are often sampled at high sampling rates to reduce the negative effects of applying low-pass filters. However, this comes at the cost of very large files that take up a lot of disk space and take a long time to process. To solve this problem, downsampling is

suggested, but a low-pass filter should be applied in order not to lose the benefits of sampling at a higher rate [10].

3.2 Artifact removal techniques

In order to remove artifacts from TMS-EEG data, there are many different techniques that can be used. In the following paragraphs, the techniques used in this work will be introduced.

3.2.1 Independent Component Analysis (ICA)

ICA is a blind source separation (BSS) technique that decomposes the EEG signal into statistically independent and non-Gaussian components (ICs). The key assumption is that each artifact has a fixed scalp distribution. Therefore, one could assume that EEG consists of the sum of the scalp distribution of the artifacts plus the scalp distribution of the other brain signals.

When applying ICA three assumptions should always be met: (a) the components should be independent, (b) the components should be non-Gaussian, (c) the number of ICs searched for should not be bigger than the number of sources, i.e., channels.

The concerns when using ICA include the fact that ICA assumptions are often not met in real data, and the algorithm will produce an output even when proper independent components do not exist [9].

3.2.2 Principal Component Analysis (PCA)

PCA is a blind source separation (BSS) technique that identifies patterns in EEG data by finding components that explain the most variance.

The method is based on two key assumptions: (a) the components are uncorrelated with each other, and (b) the first component explains as much variance as possible, with each subsequent component explaining the maximum remaining variance while being orthogonal to previous components.

PCA assumes that components with high variance represent real brain signals while those with low variance are related to noise [9].

3.2.3 Signal Space Projection (SSP)

SSP removes artifacts that have different spatial patterns from brain signals, even when they occur at the same time. A spatial pattern refers to the specific distribution of electrical activity

across the scalp electrodes - essentially showing where and how strongly signals appear at different recording sites. SSP uses PCA to identify these artifact spatial patterns and then removes those patterns from the data [9].

3.2.4 Source model based methods

Source-model-based methods are techniques that use the lead-field matrix, as well as consecutive inverse and forward modeling steps, to separate artifact signals from the data [10].

The lead-field matrix describes the relationship between neural sources in the brain and signals recorded at each EEG electrode. Essentially, it maps how activity from different brain regions contributes to measured voltages. Forward modeling uses the matrix to predict the EEG signals that would be recorded with a given source configuration. Inverse modeling works in reverse, estimating the underlying brain sources from the observed EEG data [9].

One successful example is the source-utilized noise-discarding algorithm (SOUND), which identifies and filters out signals that are unlikely to come from genuine brain activity, such as interference from electrode movement or electrical noise.

3.2.5 Discrete Wavelet Transform (DWT)

The wavelet transform is a mathematical technique used to break down a signal into its frequency components while retaining information about when those frequencies occur. A wavelet is a small, wave-like function that is localized in both time and frequency; an example is shown in Figure 3.1.

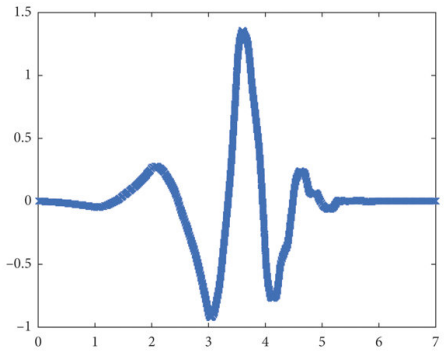


Figure 3.1 Approximation of Daubechies 4 (Db4) wavelet function [21]

The discrete wavelet transform (DWT) is the digital version of the wavelet transform. Given a chosen wavelet function and the number of decomposition levels, the DWT returns the decomposed signal. For each level, a high-pass filter and a low-pass filter, both derived

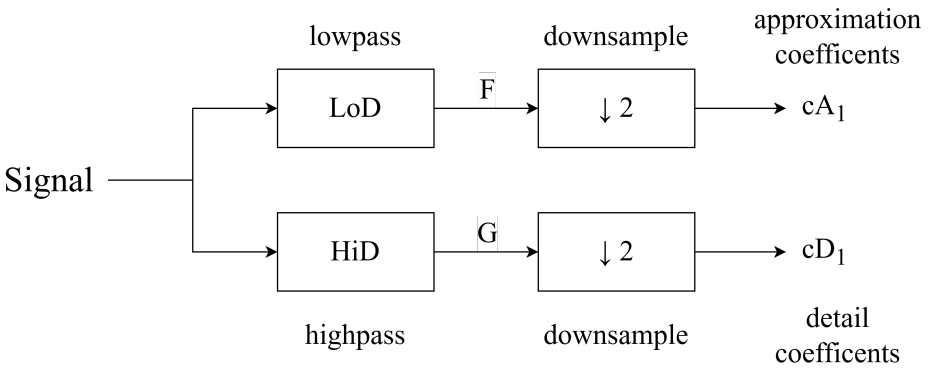


Figure 3.2 Diagram of a step of the discrete wavelet transform.

from the selected wavelet, are applied. The high-pass filter produces the detail coefficients (high-frequency content). The low-pass filter produces the approximation coefficients (low-frequency content). Then the outputs are downsampled by a factor of two (Figure 3.2). The same process is then repeated recursively on the approximation coefficients for the desired number of levels.

3.3 TMS-EEG preprocessing pipelines

A preprocessing pipeline is a set of ordered operations applied to a signal after acquisition and before analysis (Figure 3.3). This step is crucial because raw EEG data needs to be preprocessed before being analyzed. The effectiveness of the preprocessing pipeline is essential to make accurate assumptions about the data.

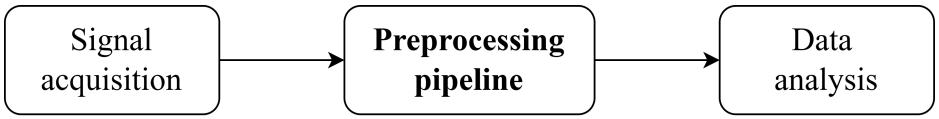


Figure 3.3 Diagram showing the placement of a preprocessing pipeline in a generic TMS-EEG data processing scenario.

The problem is that the TMS-EEG data analysis doesn't have a gold standard preprocessing pipeline [2]. In the next paragraphs, two of the most used pipelines in the domain literature will be described.

3.3.1 Two-step FastICA pipeline

The two-step FastICA approach is one of the most well-established methods for preprocessing TMS-EEG data [24, 2].

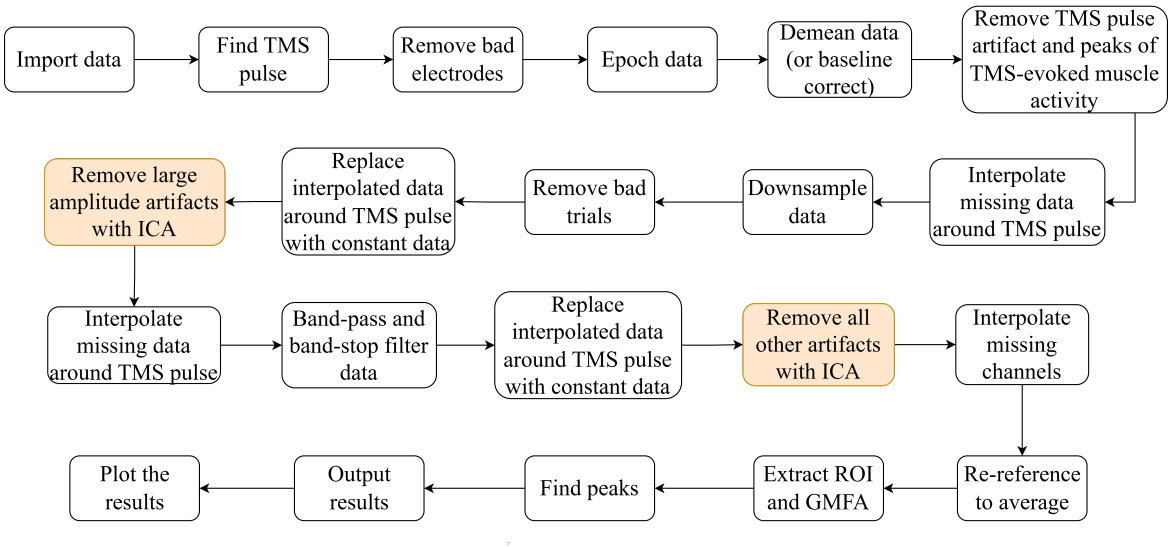


Figure 3.4 Two-step FastICA pipeline example (Rogasch et al. [24]).

It is based on the principle that large TMS artifacts can mask smaller, more subtle artifacts when ICA decomposition is first applied. The first step involves applying FastICA to identify and remove prominent artifacts, such as TMS-evoked muscle, electrical, and movement artifacts. The second step involves an additional FastICA decomposition to remove residual artifacts.

Several established TMS-EEG preprocessing pipelines and toolboxes have adopted this iterative approach, considering it more effective than single-step FastICA [24]. A pipeline example is shown in Figure 3.4.

3.3.2 SOUND/SSP-SIR pipeline

The source-utilized noise-discarding algorithm (SOUND) pipeline, incorporating signal space projection (SSP) and source-informed reconstruction (SIR), is a two-stage spatial filtering approach for preprocessing TMS-EEG data developed by Mutanen et al. [17, 18].

The SOUND step automatically detects and removes noise by iteratively comparing the trace of the channel of interest against the recordings of the other channels [16].

The SSP-SIR step suppresses various types of artifacts, like muscle artifacts and peripheral evoked potentials (PEPs), including somatosensory, auditory, and visual artifacts [16].

4

Dataset

The dataset used for the proposed experiments has been collected by Romero Lauro et al. [13]. It contains TMS-EEG recordings from 14 subjects, acquired before and after transcranial Direct Current Stimulation (tDCS). The data was recorded using sixty EEG electrodes (Figure 4.1). The data acquisition was performed by using a sample-and-hold circuit, which has the peculiarity of not recording the signal immediately after the TMS pulse (around 12 ms), to avoid the saturation of the amplifier. The sampling rate was set at 1450 Hz. It is relatively low, having that conventional TMS-EEG signal is sampled at least at 5 kHz. This is possible as the TMS pulse, which has extremely high frequency components, is not registered with the sample-and-hold recording approach.

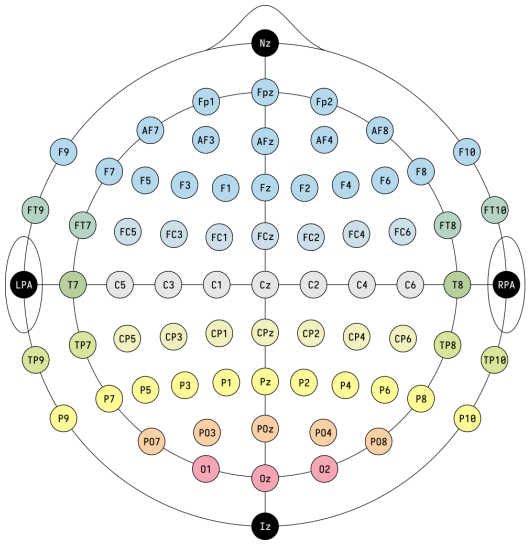


Figure 4.1 EEG electrode positions in the 10-10 system, colored as the associated lobes of the brain (Figure 2.1).*Source: Laurens R. Krol via Wikimedia Commons.*

In addition to the EEG electrodes, four auxiliary channels were included. The "GATE" channel records the time of the TMS pulse. The "EOG" channel is an electrode placed near the eyes to register ocular movements. And finally, there are two unused channels named "TRIG1" and "TRIG2". Note that the "P2" electrode was removed to allow placement of the tDCS anode.

5

Proposed solutions

This chapter describes the implementation of the artifact removal pipelines and alternative methods for ocular artifact suppression used in this study. Section 5.1 describes the structure and parameters of two widely adopted TMS-EEG preprocessing pipelines: the Two-Step FastICA and the SOUND/SSP-SIR approaches. Section 5.2 presents two alternative techniques for ocular artifact removal: Wavelet Decomposition (WD) and Empirical Mode Decomposition (EMD). The goal is to evaluate whether these methods can serve as effective substitutes for the standard ICA-based approach, especially in contexts where ICA assumptions may be violated. All methods were applied to the same dataset introduced in Chapter 4 to ensure comparability of results.

5.1 Implementation of the pipelines

In the context of TMS-EEG data analysis, the *preprocessing* phase is essential for ensuring the quality of subsequent phases. For the scope of this work, two reference pipelines from existing literature were selected: **Two-Step FastICA** and **SOUND/SSP-SIR**. These pipelines were selected because they offer complementary approaches. The two pipelines were implemented using the TMS-EEG Signal Analyzer (TESA) toolbox, which is an open-source EEGLAB MATLAB extension [16, 24]. The pipelines were designed to run without user intervention and do not require domain expertise for execution. The structure of the pipelines was inspired by the work of Brancaccio et al. [2].

In the Table 5.1 are reported the following parameters: (a) the channel location which indicates which system of reference for electrodes positioning has been used, (b) the epoching length which is used to determine how much time to keep before and after the TMS pulse when creating the epochs, (c) the baseline interval that specifies the time interval to use in each epoch to perform the baseline correction, (d) the band-pass filter range which determine the frequency range of the signal we want to keep, and (e) the downsampling value which tells how many samples to keep of the original signal. To obtain comparable results, the operations that are common to both pipelines have the same parameters.

Common parameters	
Channel location	Cap385 10-5 System
Epoching length	-1000 ms to 1000 ms
Baseline interval	-1000 ms to 0 ms
Band-pass filter	1 Hz to 48 Hz
Downsampling	725 Hz

Table 5.1 Common parameters between the pipelines

5.1.1 Two-step FastICA pipeline

The Table 5.2 illustrates the fourteen steps and the relative MATLAB functions used to implement the pipeline. Some of these steps require custom parameter tuning. The two FastICA steps (`pop_tesa_fastica`) have the following parameters to be chosen:

- *Approach*: FastICA offers two modes: deflation and symmetric. The latter was selected as it is reported to be more reliable than the deflation approach [12].
- *g (contrast function)*: it is necessary to measure the non-Gaussianity of the extracted components, both "tanh" and "gauss" functions perform equally well for TMS-EEG analysis [12]. In this case, the former was selected.
- *Stabilization*: controls whether the FastICA uses a stabilized version to detect "strokes" (i.e., when the algorithm gets stuck) and halves the learning rate. In this case, the default "off" parameter was selected.

To enable automatic artifactual components detection and removal (`pop_tesa_compselect`), for each artifact type, there are some parameters to be chosen:

- *TMS-evoked muscle activity*: "tmsMuscleWin" defines the time window to use to compute the Mean Absolute Amplitude (MAA) to be compared with the MAA of the

component. If the MAA of the window is "tmsMuscleThresh" times bigger than the MAA of the component, then a TMS-evoked muscle activity artifact is detected [22].

Actual choice: "tmsMuscleWin" = 11 to 30 ms, "tmsMuscleThresh" = 8.

- *Eye blinks*: "blinkElecs" defines two electrodes near the eyes used to compute the mean absolute z-score. If the mean absolute z-score is larger than "blinkThresh", then an eye blink artifact is detected [22].

  Actual choice: "blinkElecs" = "fp1" and "fp2", "blinkThresh" = 2.5.

- *Lateral eye movement*: "moveElecs" defines two electrodes on either side of the forehead used to compute their z-score. If the two z-scores have opposite signs, the positive one is greater than "moveThresh" and the negative is smaller than -"moveThresh", then a lateral eye movement artifact is detected [22].

  Actual choice: "moveElecs" = "F7" and "F8", "moveThresh" = 2.

- *Persistent muscle activity*: "muscleThresh" defines the minimum slope of a line best fit of the Welch Power Spectral Density (PSD). If the slope of the line is greater than the selected threshold, then a persistent muscle activity artifact is detected [5, 22].

  Actual choice: "muscleThresh" = 0.6, "muscleFreqIn" = 30 to 100 Hz.

- *Electrode noise*: "elecNoiseThresh" defines the minimum z-score of a single electrode to detect an electrode noise artifact [22].

  Actual choice: "elecNoiseThresh" = 4.

- *Sensory and other artifacts*: for this kind of artifact, there are no specific rules suitable to detect artifactual components; these artifacts should be classified manually by the user. Therefore, these artifacts are not considered in this experiment.

If the default parameters do not detect a particular artifact properly, it is possible to enable the threshold feedback option for that artifact, which will automatically calculate an appropriate threshold parameter.

	Analysis steps	**EEGLAB/TESA function**
1	Import data	`pop_loadset`
2	Load channel location	`pop_chanedit`
3	Add TMS pulse events	`pop_chanevent`
4	Remove unused electrodes	`pop_select`
5	Remove bad electrodes	`pop_rejchan`
6	Downsample	`pop_resample`
7	Epoch data around TMS pulse	`pop_epoch`
8	Demean data	`pop_rmbase`
9	Remove bad trials	`pop_joinprob`, `pop_rejepoch`
10	FastICA for TMS-evoked muscle activity	`pop_tesa_fastica`, `pop_tesa_compselect`
11	Band-pass filter	`pop_tesa_filtbutter`
12	FastICA for all other artifacts	`pop_tesa_fastica`, `pop_tesa_compselect`
13	Interpolate missing channels	`pop_interp`
14	Re-reference to average	`pop_reref`

Table 5.2 Describes the processing steps and the relative functions used to implement the 2-step FastICA pipeline [2, 24].

5.1.2 SOUND-SSP/SIR pipeline

The Table 5.3 illustrates the sixteen steps and the relative MATLAB functions used to implement the pipeline. The following steps need parameter tuning: (a) high-pass filter and FastICA for ocular artifact, (b) SOUND for detection and cleaning of noise components and interpolation of noisy channels, (c) SSP-SIR for muscle artifact removal.

As for the FastICA ocular artifacts removal, the parameters are the same as discussed

in Section 5.1.1, but only the eye movement and lateral eye movement detection should be enabled. In addition, before FastICA execution, a 0.1 Hz high-pass filter is applied to improve the performance of the ICA algorithm.

The SOUND step requires two parameters:

- *Lambda value*: also known as regularization level, it is needed as the SOUND algorithm uses minimum-norm estimation (MNE). It controls the amount of cleaning, the bigger it is more noise is removed, but the risk of over-correction increases [22].

  Actual choice: "lambdaValue" = 0.1.

- *Iterations*: "iter" parameter specifies the number of iterations used to evaluate the noise in each channel. With 30-60 channel-EEG systems, 5-10 iterations are enough to ensure convergence [18, 22].

  Actual choice: "iter" = 5.

In addition to these two mandatory parameters, a personalized lead-field matrix, derived from structural Magnetic Resonance Imaging (MRI), which incorporates the sensor positions, tissue conductivity assignments, and source space geometry, can significantly improve the source localization performance. However, this step is optional as a standard spherical lead field model can be used with acceptable results, though with reduced spatial accuracy compared to the personalized approach.

The SSP-SIR step has the following customizable parameters:

- *Art scale*: determines the method to estimate the artifacts in the PCA ("automatic", "manual", "manualConstant", or "control").

  Actual choice: "artScale" = automatic.

- *Time range*: specify the time window that contains the artifact. It is disabled in case the PCA method chosen is "automatic".

- *Principal Component*: determines the method to define the number of principal components (PCs) of the artifact to be removed. There are three options: (a) "plot" lets the user decide the number of components from a line graph that represents the dimensions of the artifact, (b) "N" let the user specify the number of components to remove, and (c) "data" to specify the percentage of variance that the components should explain.

  Actual choice: "PC" = "data", 90.

	Analysis steps	**EEGLAB/TESA function**
1	Import data	`pop_loadset`
2	Load channel location	`pop_chanedit`
3	Add TMS pulse events	`pop_chanevent`
4	Epoch data around TMS pulse	`pop_epoch`
5	Remove unused electrodes	`pop_select`
6	Remove bad electrodes	`pop_rejchan`
7	Remove bad trials	`pop_joinprob`, `pop_rejepoch`
8	Demean data	`pop_rmbase`
9	High-pass filter 0.1 Hz	`pop_tesa_filtbutter`
10	FastICA for ocular artifact removal	`pop_tesa_fastica`, `pop_tesa_compselect`
11	SOUND for detection and cleaning of noise components and interpolation of noisy channels	`pop_tesa_sound`
12	Baseline correction	`pop_rmbase`
13	SSP-SIR for muscle artifact removal	`pop_tesa_sspsir`
14	Band-pass filter	`pop_tesa_filtbutter`
15	Baseline correction	`pop_rmbase`
16	Downsample	`pop_resample`

Table 5.3 Describes the processing steps and the relative functions used to implement the SOUND/SSP-SIR pipeline [2, 17, 18].

5.2 Alternative methods for ocular artifact removal

The ocular artifacts (Section 2.3) are an important source of noise in TMS-EEG data. A domain expert or an algorithm could manually detect and remove the affected trials, but it would imply a very big data loss, because blinks occur with a frequency of 20 per minute and with a duration between 50 and 500 milliseconds [7]. Therefore, it is fundamental to have a reliable ocular artifact removal method.

The goal of this experiment is to compare the standard FastICA *ocular artifact removal* approach with two alternative methods. The two methods in question are **Wavelet decomposition (WD)** and **Empirical Mode Decomposition (EMD)**. The objective is to identify a method that preserves TEP information as much as possible while eliminating ocular artifacts. This, without violating his own assumptions, like often happens using ICA.

Given a raw TMS-EEG dataset already epoched (-1000 ms to 1000 ms) around the TMS pulse, the following pipeline is applied:

1. Remove unused channels.
2. Add EOG events to the dataset (optional).
3. Remove bad channels.
4. High-pass filter 0.1 Hz.
5. Apply ocular artifacts removal method:
   (a) FastICA.
   (b) Wavelet Decomposition.
   (c) Empirical Mode Decomposition.

The FastICA method was implemented using the same parameters that were employed in the SOUND/SSP-SIR pipeline (Section 5.1.2).

The Wavelet method was implemented using the Daubechies 4 (db4) wavelet function, as mother wavelet, because it is a commonly used wavelet function for EEG data analysis [14]. Twelve levels of decomposition were used to obtain a final approximation level with very low frequency content. Then, to remove the ocular artifact, the levels from 7 to 12, which contained most of the ocular artifact peaks, were removed.

The EMD method was implemented using Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) as an interpolation method, which is suggested for non-smooth signals like

EEG [22]. For the other parameters, the default values were chosen as no performance improvement was found by changing them. The Intrinsic Mode Functions (IMFs) removed were the third and the fourth, as they were the components containing most of the high-amplitude artifact.

The Table 5.4 summarizes the parameters chosen for each method and specifies the relative MATLAB function used.

Method	MATLAB function	Parameters	Values
FastICA	`pop_tesa_fastica` `pop_tesa_compselect`	Approach g Blink Threshold Move Threshold	symm tanh 2.5 2
Wavelet transform	`wavedec`	Wavelet type Levels of decomposition Detail levels removed	Daubechies 4 (db4) 12 7,8,9,10,11,12
Empirical Mode Decomposition	`emd`	Interpolation Sift Relative Tolerance Max Num IMF Sift Max Iterations Max Number Extrema Max Energy Ratio IMFs removed	pchip 0.2 5 100 1 20 3,4

Table 5.4 Functions and parameters used in the three ocular artifact removal methods.

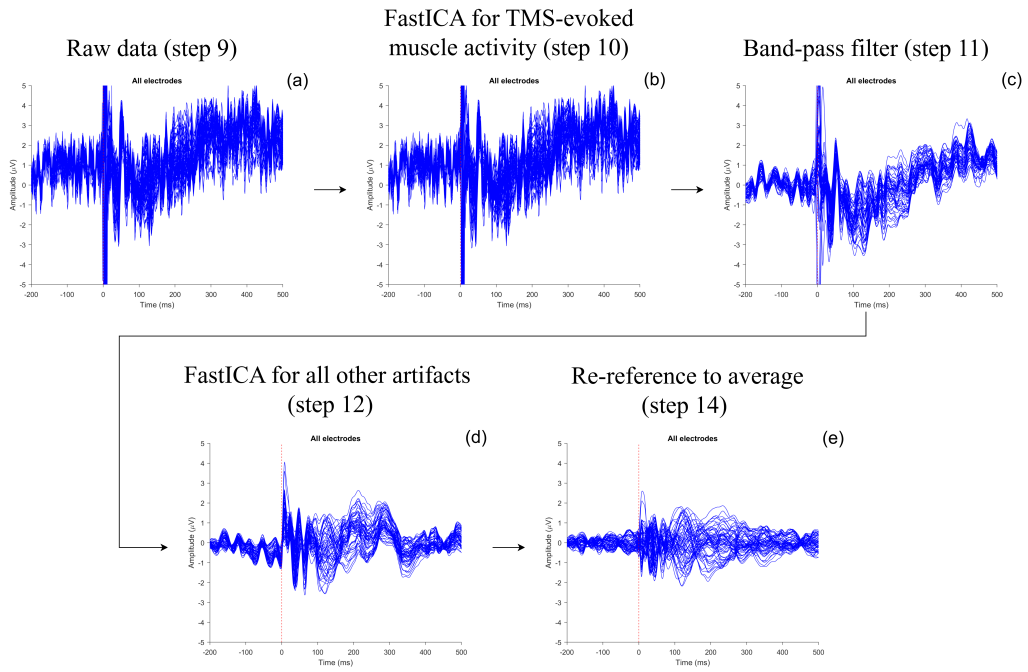
6
Results

This chapter presents the results obtained by the pipelines and the ocular artifact removal methods presented in Chapter 5. In particular the Section 6.1 compares the Two-step FastICA and SOUND/SSP-SIR pipelines, by evaluating their impact on signal integrity and artifact removal. The Section 6.2 instead deepens the comparison between three different methods for ocular artifact removal (FastICA, Wavelet, and EMD) with a qualitative and quantitative analysis.

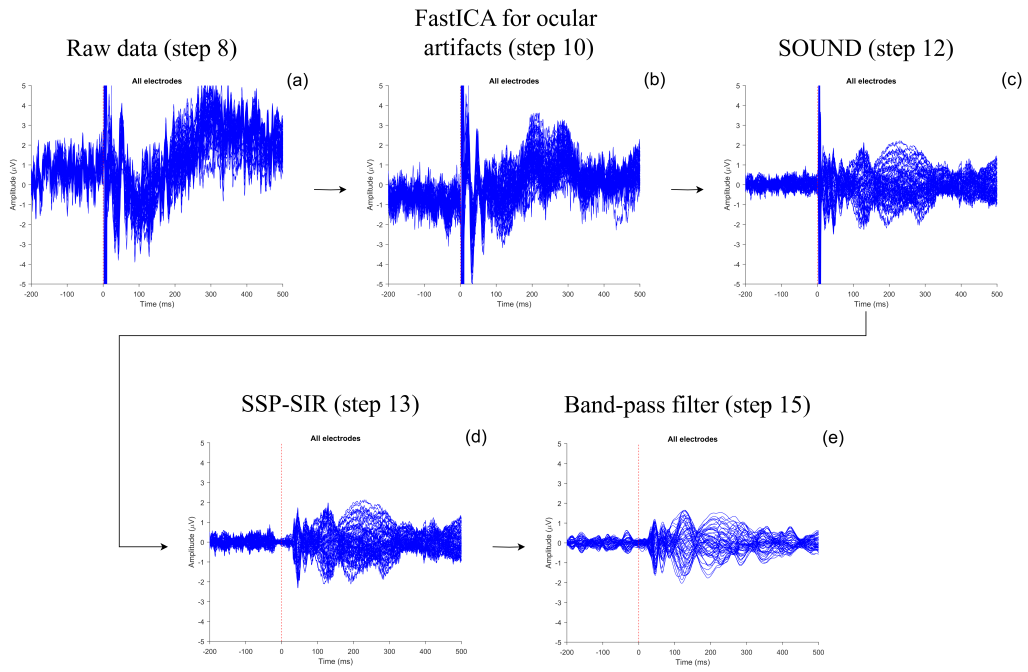
6.1 Comparison of the pipelines

In this section, the results of the two pipelines described in Section 5.1 are compared: Two-step FastICA and SOUND/SSP-SIR. Both pipelines are applied to the same dataset (Chapter 4), and every significant elaboration step has been presented using the TEP visualization. The target is to analyze how each pipeline phase modifies the signal in terms of artifact cleaning and neural signal preservation. The Figure 6.1 follows the order of the operations described by Table 5.2 and Table 5.3.
The Two-step FastICA pipeline marked "P2" and "AFz" electrodes for rejection, and removed 3 bad trials (i.e. epochs). The first FastICA step removed only one component (IC), which explains why in Figure 6.1a the plot in panel (a) and panel (b) are very similar. Instead, the TEP in panel (c) is very different as the band-pass filter removes the high-frequency components from the signal; this step is possible as normal brain activity is lower than 50 Hz (Section 2.1). Then, the graph in panel (d) shows the application of the second FastICA step,

which has a bigger impact than the precedent ICA step. This is particularly visible around time zero, where, compared to the previous plot, the signal amplitude is attenuated. Finally, graph (e) provides a clean visualization of the signal by re-referencing it.



(a) Two-step FastICA pipeline.



(b) SOUND/SSP-SIR pipeline.

Figure 6.1 Two-step FastICA and SOUND/SSP-SIR pipelines mean TEPs visualization after each significant elaboration step. The relative steps are reported in Table 5.2 and Table 5.3.

The SOUND/SSP-SIR pipeline also marked "P2" and "AFz" electrodes for rejection, but removed only 2 bad trials. The FastICA for ocular artifacts (OAs) removal step marked four components as OAs and removed them. Compared to the Two-step FastICA pipeline, the second TEPs' plot (b) in Figure 6.1b presents a bigger difference in terms of TEPs shape, from the previous in panel (a). The graph in panel (c) reveals a big difference in terms of amplitude and removes the artifactual shape of the signal from 0ms to 80ms. Then, the SSP-SIR step removes the high amplitude artifact after the TMS-pulse. Finally, graph (e) shows the effect of the band-pass filter in removing high frequency components and baseline corrections.

To evaluate how each pipeline modifies the signal's frequency spectrum, Figure 6.2 shows the Power Spectral Density (PSD) of the signal before processing and after each pipeline application. The two pipelines have a similar effect on the signal in terms of high frequency components. However, the SOUND/SSP-SIR pipeline removes more low frequency signal between 0 and 10 Hz, compared to the Two-step FastICA. A noisy shape of the SOUND/SSP-SIR signal should also be noted in the low-frequencies.

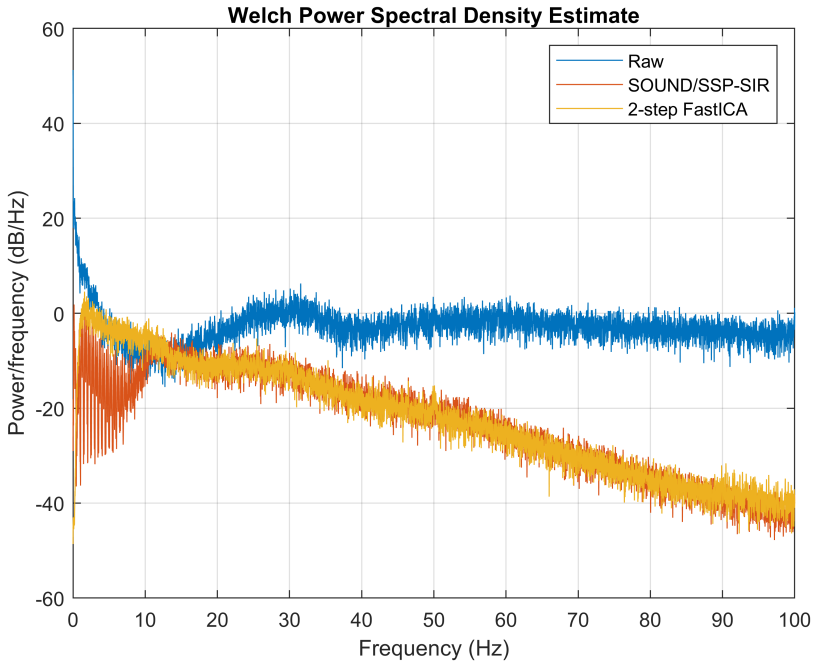


Figure 6.2 Power Spectral Density of the original and the two elaborated signals (FP1 channel).

6.2 Comparison of methods for ocular artifact removal

In this section, the results of three different approaches for the removal of Ocular Artifacts (OAs) are analyzed and compared: **Fast Independent Component Analysis (FastICA)**, **Wavelet Decomposition (WD)**, and **Empirical Mode Decomposition (EMD)**. Each method was applied separately to the same EEG signal, and its effectiveness was evaluated in two domains: in the *temporal domain*, by analyzing a signal segment containing an identified ocular artifact, and in the *frequency domain*, through spectral analysis to assess how each method affected the power distribution across frequency bands.

To objectively evaluate the performance of each denoising method, it is important to identify the timing of the artifact in the raw EEG signal, to be able to understand if the artifact has been removed from the signal after the processing. In this case has been used the **electrooculography (EOG)** electrode, which is an electrode placed near the eyes, to detect eye blinks and eye movements accurately. To identify the timing of the events, a simple peak detection algorithm was applied to the EOG signal. The detection threshold was computed using the following heuristic formula: $threshold = mean(EOG) + 1.2 * std(EOG)$. This thresholding rule is a simple yet effective heuristic based on the assumption that eye blinks produce transient deflections significantly higher than baseline fluctuations. All the peaks above this threshold and distant at least 1 second from each other were marked as EOG artifacts. This minimum time interval avoids marking multiple peaks caused by the same blink. The result is displayed in Figure 6.3. It is important to note that this approach relies on the availability of a reference EOG channel.

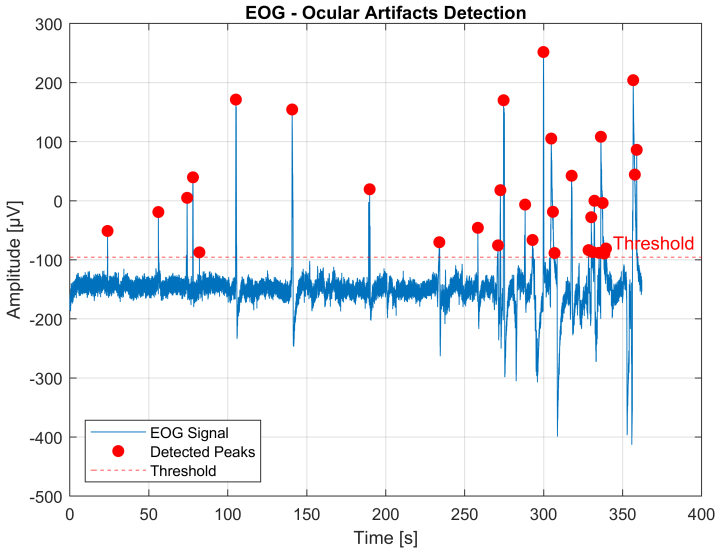
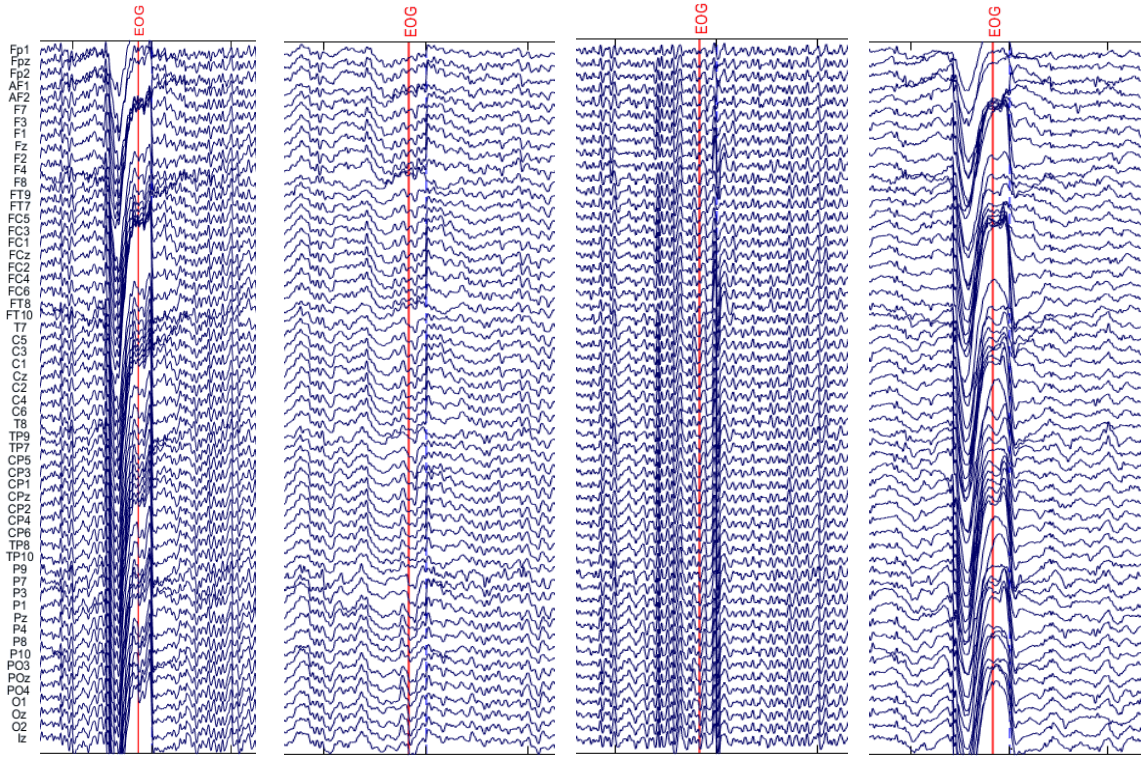


Figure 6.3 Electrooculogram signal used to find ocular artifacts.

To understand concretely what an ocular artifact is and what it means to remove it, in Figure 6.4 are presented four EEG signal plots, on 60 channels, between 77 ms and 79 ms after the beginning of the recording. In that time range, an EOG artifact was found by the previous algorithm. The first signal in panel (a) displays the raw EEG signal with a prominent ocular artifact in the middle, which affects all the channels. The second image (b) shows the EEG signal after the FastICA ocular artifact removal, where the ocular artifact has been massively reduced, but a small artifact is still present just after the red vertical line representing an EOG event. Also in the third image (c), where the wavelet method has been applied, the ocular artifact has been broadly mitigated, but some artifactual shapes are still visible before and after the EOG event. Finally, the fourth image (d) shows the application of the EMD method, which has not been able to separate and remove the Ocular artifact properly, and still presents the ocular artifact.



(a) Raw data. (b) FastICA. (c) Wavelet. (d) EMD.

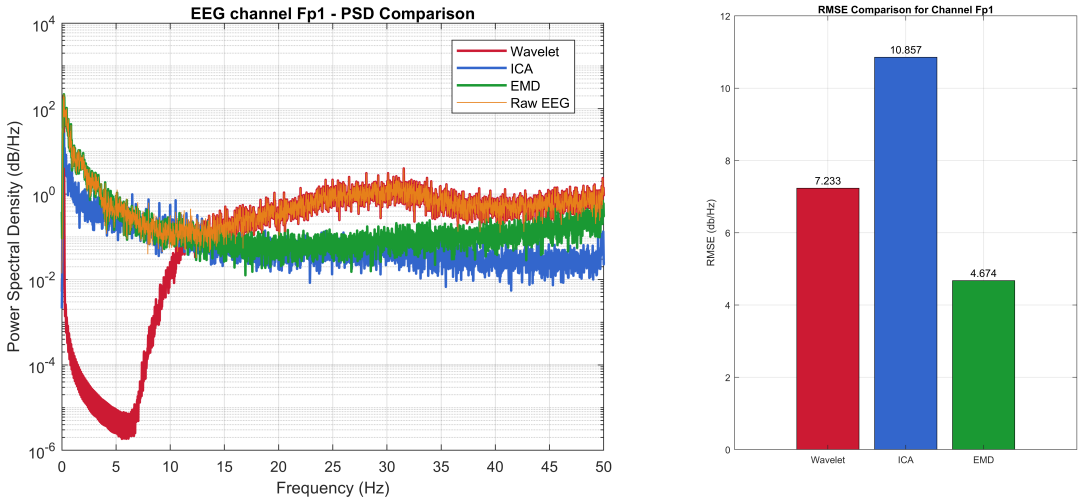
Figure 6.4 Comparison of EEG channels (epoch 39-40): (a) raw, (b) after fastICA, (c) after wavelet decomposition, (d) after Empirical Mode Decomposition. The scale used in the MATLAB EEGLAB "eegplot" function is 13 μV.

To analyze the effects on the frequency domain of the signal, Figure 6.5a shows the Power Spectral Density of the raw signal and the three processed signals, from 0 to 50 Hz. The graph shows that the wavelet (red line) is the method that removed the most low frequencies

(0 to 13 Hz), while the ICA method (blue line) was more conservative and the EMD method (green line) did not remove low frequencies at all, as it is overlapped with the raw signal (yellow line). Instead, with the medium and higher frequencies (13 to 50 Hz), the trend changed. The wavelet processed signal is overlapped with the raw signal, while the ICA and the EMD signals are lower. This means that the wavelet method has been more conservative with medium to high frequencies compared to EMD and ICA.

Considering that electrooculographic (EOG) activity is in the range between 0 and 12 Hz [7], it is clear why Wavelet and ICA perform better than the EMD method. But neither ICA nor wavelet methods are perfect. The wavelet method seems too aggressive on low frequencies, while the FastICA algorithm affects frequencies that do not contain ocular artifacts. It should be noted that FastICA could have performed better if the independent components classification had been supervised by a domain expert, but this is not the purpose of the experiment.

Then, Figure 6.5b provides a quantitative analysis of the PSD by displaying the Root Mean Square Error (RMSE) computed between the original raw signal and each processed signal. The RMSE highlights that the EMD signal is the most similar in term of frequencies (i.e. most conservative), the ICA signal is the most different and the Wavelet signal is in the middle between the two.



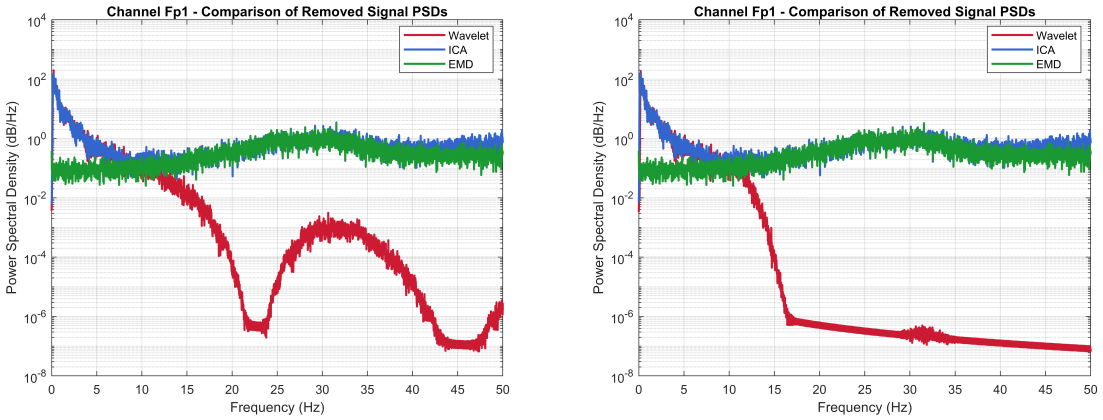
(a) PSD

(b) RMSE

Figure 6.5 Frequency analysis of the signals: (a) Power Spectral Density (PSD) EEG channel Fp1 after OAs removal using Wavelet, ICA, and EMD methods. With the raw EEG signal as a reference. (b) Root Mean Square Error (RMSE) values for Wavelet, ICA, and EMD methods compared to the raw EEG signal for channel Fp1.

Finally, Figure 6.6 shows the power spectral density (PSD) of the removed signal from each pipeline. The signal removed by the wavelet method in Figure 6.6a shows an artifactual, wavy shape caused by the signal's reconstruction. Using a wavelet with more coefficients, such as Daubechies 20 (db20), reduces the artifact. The result of using db20 is shown in Figure 6.6b.

The problem with using a higher-order wavelet function, such as db20, is that the number of coefficients of the mother wavelet grows, and with that, the computational cost increases. Notice that while the filtering artifact is reduced, it is imperceptible in terms of the final signal frequency content. The artifact in Figure 6.6a does not really affect the processed signal as the power of the frequencies is very low.



(a) Wavelet function: db4.

(b) Wavelet function: db20.

Figure 6.6 Comparison of removed signal power spectral densities (PSDs) for Channel Fp1 using different artifact removal techniques (Wavelet, ICA, and EMD) with two wavelet functions: (a) db4 and (b) db20.

7 Discussion

7.1 Which is the best pipeline?

The biggest problem in determining the objective performance of a TMS-EEG preprocessing pipeline is the impossibility of recovering the *ground-truth* signal, which is the pure signal produced exclusively by the brain. For this reason, it is very difficult to objectively determine which pipeline has performed better between the Two-step FastICA and the SOUND/SSP-SIR. Some works in the domain literature make a quantitative performance analysis by creating a synthetic signal [2], which is done by considering a signal as ground-truth and then manually adding the artifacts. However, until a benchmarking dataset is developed, it will be difficult to validate a pipeline [23].

Furthermore, these difficulties are enhanced by the different tools and techniques used to record the TMS-EEG signal worldwide. For example, the dataset used for this experiment was recorded using a sample-and-hold circuit, but nowadays newer amplifiers record the TMS-pulse artifact. This results in the necessity of tuning the chosen pipeline according to the way the signal has been recorded.

In conclusion, the best pipeline cannot be determined objectively without a ground-truth signal, but based on the results can be chosen the best fitting pipeline for the purpose of the experiment.

7.2 Are the alternative methods good solutions?

The Wavelet Decomposition (WD) and the Empirical Mode Decomposition (EMD) methods applied to ocular artifacts removal have performed differently.

The wavelet method managed to remove the ocular artifact by preserving the higher frequencies. However, the problem is that it has been too aggressive in removing low-frequency brain signals. It could be a promising method to use in case the low-frequency brain waves like alpha waves (Section 2.1) are not of interest for the specific experiment.

On the contrary, the EMD method failed to remove the ocular artifact, suggesting that in order to be used for this purpose it needs a more sophisticated approach.

The FastICA ocular artifact removal method is the standard method to remove ocular artifacts in the domain literature. It performed very well in removing the ocular artifact while preserving low frequencies. However, it also removed frequencies that are not related to ocular artifacts, which means it also removed brain signals. As said, this could improve by manually selecting the components to remove, but while aiming to preserve the brain signal, the ocular artifact removal could get worse. This can happen because ICA assumptions are often violated in real EEG data; components may appear independent mathematically, but still contain mixtures of brain and ocular signals, making separation imperfect.

7.3 Future developments

Following this work, several potential directions have emerged for improving and extending the current artifact removal approaches.

Two-step FastICA and SOUND/SSP-SIR pipelines

1. Explore the use of the `feedback` option in the `pop_tesa_compselect` function to obtain more accurate and data-driven thresholding parameters for component rejection.

2. Develop an automatic classification system for FastICA components, capable of distinguishing between ocular, sensory, and other non-neural artifacts without requiring manual intervention.

3. Investigate the effect of different high-pass filter settings before FastICA in the SOUND/SSP-SIR pipeline. In particular, the currently used 0.1 Hz high-pass filter may not be optimal in terms of signal-to-noise ratio (SNR), as suggested by Winkler et al. [29].

4. Design an automatic parameter tuning algorithm to adaptively optimize preprocessing settings for each pipeline based on signal characteristics or artifact profiles.

Ocular artifact removal methods

1. To reduce the impact of the wavelet method on low-frequency neural components, consider replacing the full removal of entire coefficient levels with a more selective thresholding approach that suppresses only the artifact peaks while preserving underlying signal activity.

2. Test the use of biorthogonal wavelet functions, such as Biorthogonal 2.6 (bior2.6), which may improve artifact suppression due to their symmetric filters and linear phase properties, preserving signal morphology while reducing distortion [20].

3. Since the current ocular artifact detection algorithm depends on the presence of an EOG reference channel, future work could explore alternative solutions, such as machine learning classifiers trained to detect ocular artifacts directly from EEG data in the absence of EOG.

8

Conclusions

The combination of TMS and EEG techniques represents a strong and promising method to explore brain dynamics. However, its full efficacy relies on a good artifact removal process.

In this work have been implemented and compared two of the most common preprocessing pipelines in literature, the Two-Step FastICA and the SOUND/SSP-SIR pipeline. Even though they are based on two very different techniques, the results were qualitatively similar, with the Two-step FastICA being more conservative on low-frequency components.

Then was analyzed the behavior of three different ocular artifact removal methods. The literature standard FastICA method, and two proposed methods based on Wavelet Decomposition (WD) and Empirical Mode Decomposition (EMD). The results confirmed the good performance of the standard FastICA method, while exposing its critical issues, like the necessity of a domain expert for component validation or the violation of ICA assumptions. Between the Wavelet Decomposition method and the Empirical Mode Decomposition method, the WD performed better in particular has been very good in preserving the high-freqeuncy components, while the EMD method failed to remove the ocular artifacts.

In conclusion, this work highlights how the choice of an appropriate preprocessing pipeline or ocular artifact removal method must be guided by the characteristics of the dataset and the goals of the analysis. Future work could focus on developing a proper automatic definition of the pipeline parameters and a deeper analysis of ocular artifact removal using time-frequency methods.

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